

OFFICIAL TECHNICAL SPECIFICATION

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Aarogyam (MedAdhere v2.1)
Intelligent Medication Adherence Monitoring & Autonomous Intervention System

Document Identifier	SRS-AAROGYAM-2026-V2.1
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This document contains proprietary information regarding the cyber-physical design, firmware architecture, agent handover protocols, and machine learning inference pipelines of Aarogyam.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) provides an exhaustive, authoritative description of the **Aarogyam (MedAdhere v2.1)** platform. It defines all functional and non-functional requirements, external interfaces, hardware-firmware integration parameters, autonomous agent behaviors, data models, and architectural boundaries.

This document serves as the foundational reference for software engineers, firmware developers, AI/ML engineers, clinical auditors, and hardware designers collaborating on the Aarogyam ecosystem.

1.2 Scope of the System

Aarogyam is an intelligent, cyber-physical healthcare platform designed to combat the chronic disease medication non-adherence crisis in India and emerging economies. The platform integrates:

- IoT Edge Hardware:** An ESP32 dual-core smart pill dispenser equipped with stepper motor carousel rotation, differential gravimetric verification (load cell), optical/ultrasonic hand detection, physical servo-locking gates, bilingual OLED display, DS3231 RTC, and audio guidance.
- Central Cloud Backend:** A high-throughput Django 5.x / Daphne ASGI distributed service orchestrating 27 domain apps, WebSockets via Django Channels, Celery Beat task scheduling, and Redis caching.
- Agent Handover Orchestrator:** An asynchronous, event-driven service mesh coordinating 16 core agents and 12 extension agents with typed handover contracts.
- Predictive AI Engine:** Machine learning pipeline featuring XGBoost risk scoring, SHAP TreeExplainer feature attribution, and predictive refill consumption forecasting.
- Autonomous Agent Runtime:** An LLM-powered (Groq / Llama-3 / Mixtral) observe-reason-plan-act-evaluate loop executing high-risk clinical interventions and automated pharmacy replenishment with human-in-the-loop safeguards.
- Multi-Role Web Portals:** Five dedicated web dashboards (Patient, Caregiver, Doctor, Hospital Tenant Admin, and Super-Admin) developed in React 19, TailwindCSS, and TanStack Query.
- Omnichannel Tele-Intervention:** Push notifications (FCM), Telegram interactive bot, SMS/Voice (Twilio), Email (SendGrid), and WebSocket push alerts.
- Healthcare Interoperability:** Indian National Digital Health Mission (ABDM / ABHA M1/M2/M3), HL7 FHIR R4 export, OpenFDA pharmacovigilance, and HIPAA/DISHA-compliant audit trails.

1.3 Real-World Problem Statement

According to the World Health Organization (WHO), adherence to long-term therapy for chronic illnesses in developing nations averages only 50%. In India, where hypertension, diabetes, and cardiovascular diseases affect over 250 million citizens:

- Patients miss doses due to forgetfulness, cognitive decline, complex multi-drug schedules, and asymptomatic conditions.
- Caregivers and family members suffer from "alert blindness" and lack physical proof of ingestion.
- Doctors have zero visibility into actual patient compliance between quarterly visits, leading to incorrect dosage escalation (pseudo-resistance).
- Pharmacies face irregular refill patterns, resulting in stockouts and therapy discontinuation.

Aarogyam resolves this by bridging the cyber-physical gap: physically locking medicine, validating dose extraction via precision weight deltas, proactively predicting non-adherence through machine learning, and mobilizing caregivers and autonomous agents

before health emergencies manifest.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
ABHA	Ayushman Bharat Health Account (India's national digital health identifier).
ABDM	Ayushman Bharat Digital Mission (National digital health ecosystem standards).
ASGI	Asynchronous Server Gateway Interface (enables asynchronous Python web servers).
Celery Beat	Periodic job scheduler for background task queues.
DISHA	Digital Information Security in Healthcare Act (Indian healthcare privacy standard).
FHIR	Fast Healthcare Interoperability Resources (HL7 standard for electronic healthcare records).
FCM	Firebase Cloud Messaging (Google mobile push notification infrastructure).
GPIO	General Purpose Input/Output (pins on ESP32 microcontroller).
HX711	24-bit precision analog-to-digital converter designed for weigh scales.
ICD-10	International Statistical Classification of Diseases and Related Health Problems (10th Revision).
MFA / TOTP	Multi-Factor Authentication / Time-based One-Time Password.
RBAC	Role-Based Access Control.
RTOS	Real-Time Operating System (specifically FreeRTOS on ESP32).
SHAP	SHapley Additive exPlanations (explainable AI methodology).
ULN2003	High-voltage, high-current Darlington transistor array used to drive stepper motors.
WSS	WebSocket Secure protocol.

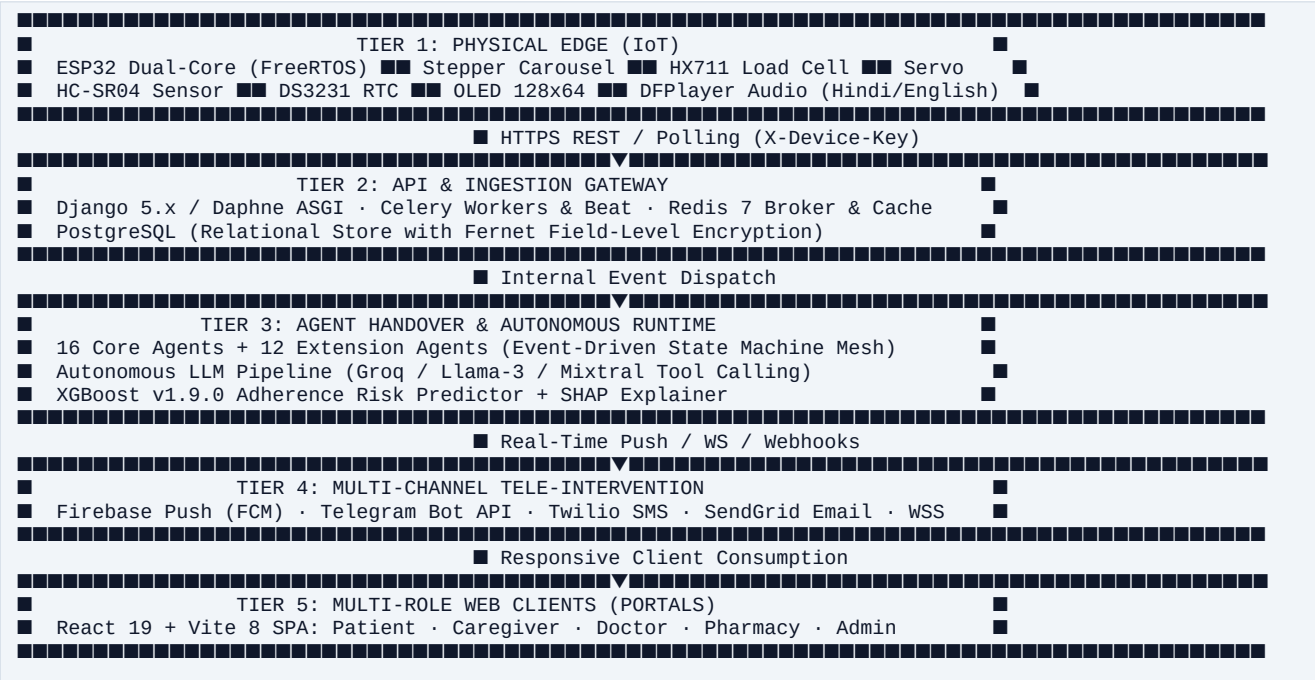
1.5 Document Conventions

- **Requirement Tags:** [FR-xxx] for Functional Requirements, [NFR-xxx] for Non-Functional Requirements, [HWR-xxx] for Hardware Requirements.
- **Priority Ratings:** High (Critical path, MVP), Medium (Core operational feature), Low (Extended capability).
- **Standards Format:** Requirements are structured with Preconditions, Triggers, Inputs, Processing Logic, Outputs, and Error Handling.

2. Overall Description

2.1 Product Perspective & Context Architecture

The Aarogyam platform operates as an integrated multi-tier cyber-physical system:



2.2 Hardware Subsystem Specification (ESP32 Smart Dispenser)

The physical dispenser is an intelligent medical appliance placed in the patient's residence.

2.2.1 Component Specifications

- 1. **Processing Core:** Espressif ESP32-WROOM-32 (Tensilica Xtensa Dual-Core 32-bit LX6, 240 MHz, 520 KB SRAM, 4 MB Flash, 802.11 b/g/n Wi-Fi, Bluetooth 4.2 BLE).
- 2. **Carousel Actuation:** 28BYJ-48 5V Unipolar Stepper Motor driven by ULN2003 Darlington Array (64:1 reduction ratio, 4096 half-steps per 360° rotation). Drives a 4-compartment rotatable pill carousel.
- 3. **Gravimetric Verification:** 1 kg Aluminum Single-Point Load Cell coupled with an Avia Semiconductor HX711 24-bit ADC differential amplifier. Resolution: ±0.05 g. Scale factor calibrated to 2280.0.
- 4. **Dispense Gate Barrier:** TowerPro SG90 9g Micro Servo Motor (PWM controlled, 50 Hz, 0° closed barrier, 90° open chute).
- 5. **Proximity & Hand Detection:** HC-SR04 Ultrasonic Ranging Module (Echo/Trigger cycle, operating threshold ≤ 15 cm, sample debounce 500 ms).
- 6. **Local Visual Interface:** \$0.96\$ I2C SSD1306 OLED Display (\$128 \times 64\$ monochrome pixels, I2C address 0x3C).
- 7. **Offline Real-Time Clock:** Maxim Integrated DS3231 Extreme Precision I2C RTC with integrated temperature-compensated crystal oscillator (TCXO) and CR2032 coin cell backup (I2C address 0x68).
- 8. **Audio Guidance Subsystem:** DFPlayer Mini MP3 Player Module driven over UART2 (16/17), paired with an 8 Ω / 3W high-fidelity miniature speaker.
- 9. **Power Subsystem:** 5V / 2A micro-USB/DC barrel input with 3.3V LDO regulator and GPIO 34 analog voltage divider for battery health telemetry.

2.2.2 ESP32 Pin Mapping Table

ESP32 Pin	Connected Subsystem	Pin Function	Electrical Interface
GPIO 13	ULN2003 Stepper Driver	IN1 (Phase A)	3.3V Logic OUT
GPIO 12	ULN2003 Stepper Driver	IN2 (Phase B)	3.3V Logic OUT

GPIO 14	ULN2003 Stepper Driver	IN3 (Phase C)	3.3V Logic OUT
GPIO 27	ULN2003 Stepper Driver	IN4 (Phase D)	3.3V Logic OUT
GPIO 25	SG90 Servo Motor	PWM Signal	50 Hz PWM OUT ($0^\circ - 90^\circ$)
GPIO 26	HC-SR04 Ultrasonic	Trigger	10 μ s Pulse OUT
GPIO 33	HC-SR04 Ultrasonic	Echo	Level-Shifted Pulse IN
GPIO 35	HX711 Load Cell ADC	DOUT	Synchronous Serial Data IN (Input Only)
GPIO 18	HX711 Load Cell ADC	SCK	Serial Clock OUT (> 100 kHz)
GPIO 16	DFPlayer Mini MP3	RX2 (ESP32 RX)	UART Serial IN (9600 bps)
GPIO 17	DFPlayer Mini MP3	TX2 (ESP32 TX)	UART Serial OUT (Ω protected)
GPIO 15	DFPlayer Mini MP3	BUSY	Active LOW Detection IN
GPIO 21	Shared I2C Bus	SDA	I2C Data (400 kHz Fast Mode)
GPIO 22	Shared I2C Bus	SCL	I2C Clock (400 kHz Fast Mode)
GPIO 32	Active Buzzer	Piezo Driver	High-Drive Logic OUT
GPIO 2	Onboard Blue LED	Status Beacon	Logic OUT
GPIO 34	Battery Voltage Divider	ADC1_CH6	Analog Input (0-3.3V, Input Only)

2.2.3 Dual-Core FreeRTOS Task Architecture

The ESP32 firmware distributes workloads across dual cores using FreeRTOS primitives:

- Core 0 — taskNetwork (Priority 1, 8 KB Stack):

- Wi-Fi watchdog with auto-reconnect and exponential backoff (15 s).
- NTP time synchronization updating DS3231 RTC registers.
- Heartbeat POST every 5 min delivering battery level, Wi-Fi RSSI, and uptime.
- Schedule poll GET every 60 s retrieving upcoming dosage events.
- Command poll GET every 10 s checking for ROTATE, PLAY_VOICE_NOTE, LOCK_LID, UNLOCK_LID, FILL_MODE, SYNC_TIME, and RESET_FLAGS.

- Core 1 — taskMotor (Priority 3, Highest, 4 KB Stack):

- Consumes commands from FreeRTOS xMotorCmdQueue.
- Acquires xMotorMutex to prevent concurrent stepper rotation.
- Executes precise 4096-half-step sequence to rotate compartment 1 through 4.
- Publishes COMPARTMENT_ROTATED event flags upon completion.

- Core 1 — taskSensor (Priority 2, 6 KB Stack):

- Executes real-time dispenser state machine: STATE_IDLE $\rightarrow$ STATE_DISPENSING $\rightarrow$ STATE_GATE_OPEN $\rightarrow$ STATE_WEIGHT_CHECK $\rightarrow$ STATE_GATE_LOCKED $\rightarrow$ STATE_FILL_MODE.

2.3 Software Monorepo Structure

The backend is structured into 27 discrete Django applications managed under backend/apps/:

1. core: Abstract timestamped models, shared exceptions, UUID primary keys, and cryptography helpers.
2. identity: Custom User model, JWT token generation, role assignment, TOTP MFA, session management, and UserDevice

push tokens.

3. **clinical**: ICD-10 disease taxonomy, patient clinical histories, contraindications, and clinical encounter notes.
4. **scheduling**: Prescription models, daily reminder rule generation, dosage calculation, and dose event logs.
5. **telemetry**: High-frequency IoT metrics ingest (RSSI, voltage, sensor jitter), and device health monitoring.
6. **iot**: Physical device registration, pairing tokens, firmware heartbeat handler, and command dispatch queues.
7. **subscriptions**: Tiered billing models (Free, Caregiver Plus, Family Pro, Hospital Enterprise) and payment gateway webhooks.
8. **store**: E-commerce catalog for hardware dispensers, replacement carousels, and order fulfillment.
9. **notifications**: Multi-channel notification pipeline (In-App, FCM Push, Telegram Bot, Twilio SMS/WhatsApp, SendGrid Email).
10. **communications**: WebRTC signaling, real-time WebSocket chat and video consultation between doctors and patients.
11. **audit**: Immutable, write-only audit trail logging all PHI access for HIPAA and DISHA compliance.
12. **admin_panel**: Platform-wide administrative oversight, system health diagnostics, and user moderation.
13. **analytics**: Adherence score aggregation, cohort analytics, adherence timelines, and CSV/PDF export.
14. **ai_engine**: Machine learning inference service hosting the XGBoost classifier, SHAP explainability, and refill consumption forecast.
15. **agent_runtime**: Autonomous agentic pipeline (observe, reason, plan, act, evaluate) utilizing Groq LLM tool calling for high-risk adherence and refill resolution.
16. **pharmacy**: Pharmacy partner registry, API bridge, auto-refill triggers, and inventory tracking.
17. **doctor_portal**: Doctor profile management, doctor-patient links, digital prescription generation, and risk alerts.
18. **telegram_bot**: Full-duplex interactive Telegram bot service replacing legacy chatbots with inline keyboard dose logging and OTP onboarding.
19. **family**: Family groups, multi-patient guardian linking, and consolidated family adherence dashboards.
20. **fhir_integration**: HL7 FHIR R4 standard serialization and patient record import/export.
21. **vitals**: Patient vital signs tracking (blood pressure, fasting/post-prandial blood glucose, pulse, SpO2, weight, body temperature).
22. **gamification**: Adherence streaks, badge awarding, and behavioral reward points.
23. **pharmacovigilance**: Automated drug-drug and drug-food interaction checks against OpenFDA APIs and adverse drug reaction reporting.
24. **insurance_reports**: Automated PDF generation of verified adherence certificates for health insurance premium discounts.
25. **geofence**: Patient geofence boundary configuration and location-triggered medication alerts.
26. **abha**: Ayushman Bharat Digital Mission (ABDM) integration (ABHA creation, verification, and consent management).
27. **tenants**: Multi-tenant hospital/clinic isolation with custom domain routing and isolated patient populations.

2.4 User Classes and Detailed Personas

1. Patient (End User):

2. *Profile*: Elderly individuals, chronic illness patients (diabetes, cardiovascular disease, hypertension, tuberculosis) managing 2–6 daily medications.
3. *Needs*: Frictionless reminders, physical access control, voice instructions in native language (Hindi/English), visual OLED schedule indicators, zero complex device setups.

4. Caregiver (Family Member / Guardian):

5. *Profile*: Son, daughter, or designated nurse living with or away from the patient.
6. *Needs*: Instant alerts when doses are missed or dispenser is tampered with; remote lid lock/unlock controls; weekly adherence trend summaries; emergency geofencing.

7. Doctor / Healthcare Provider:

8. *Profile*: Treating physicians, cardiologists, and endocrinologists prescribing chronic therapy.
9. *Needs*: Verifiable, tamper-proof adherence records; explainable AI risk indicators; clinical prescription builder with drug interaction checks; patient risk alerts.

10. Pharmacist / Fulfillment Partner:

11. *Profile*: In-house or partnered retail pharmacy dispensaries.
12. *Needs*: Predictive refill orders generated 5 days before medication exhaustion; prescription verification; order fulfillment status tracking.

13. Hospital Tenant Administrator:

14. *Profile*: Clinical operations manager at hospitals utilizing Aarogyam for outpatient monitoring.

15. *Needs*: Departmental patient analytics, staff doctor assignments, tenant branding, subscription billing, and compliance auditing.
16. **System Super-Administrator**:
17. *Profile*: Platform technical operators and site reliability engineers.
18. *Needs*: System health diagnostics, global audit logs, AI model performance monitoring, IoT device fleet provisioning, and API rate-limit controls.

3. External Interface Requirements

3.1 User Interfaces (Frontend Web Portals)

The web user interface is built as a single-page application (SPA) using React 19, Vite 8, TailwindCSS 4, and TanStack Query 5. The interface provides 5 role-based dashboards:

3.1.1 Patient Portal (/patient/*)

- **Dashboard (/patient/home)**: Current medication schedule, next dose countdown, daily adherence streak counter, quick dose confirmation, and recent AI insights.
- **My Medications (/patient/medications)**: Active prescriptions, dosage timings, pill images, remaining tablet counts, and refill requests.
- **IoT Dispenser (/patient/dispenser)**: Live dispenser connection state, battery level, Wi-Fi signal strength, and compartment filling wizard.
- **Vitals Tracking (/patient/vitals)**: Interactive charts for blood glucose and blood pressure with target thresholds.
- **Gamification (/patient/rewards)**: Earned badges, current streak days, and adherence level progression.

3.1.2 Caregiver Portal (/caregiver/*)

- **Patient Overview (/caregiver/dashboard)**: Multi-patient card grid displaying adherence percentage, last recorded dose, and current dispenser status.
- **Patient Detail (/caregiver/patient/:id)**: Comprehensive medical record, adherence calendar, missed dose timelines, and real-time alerts.
- **Remote Dispenser Controls**: One-click remote gate unlock, emergency dispense trigger, and dispenser lock override.
- **Geofencing Config (/caregiver/patient/:id/geofence)**: Interactive Leaflet map to establish home radius and safe travel zones.

3.1.3 Doctor Portal (/doctor/*)

- **Clinical Dashboard (/doctor/home)**: High-risk patient triage queue flagged by the AI engine, pending refill requests, and patient alerts.
- **Prescription Builder (/doctor/patient/:id/prescribe)**: Digital prescription interface with automated OpenFDA drug interaction warnings and auto-calculated schedule generation.
- **Adherence Analytics**: Detailed dose timing variance, missed dose heatmaps, and exportable clinical PDF reports.

3.1.4 Hospital Tenant Admin & Super-Admin (/admin/*)

- **Tenant Management (/admin/tenants)**: Multi-tenant clinic onboarding, domain mapping, doctor assignments, and subscription license allocation.
- **IoT Device Fleet (/admin/devices)**: Batch device ID generator, hardware firmware version tracking, and diagnostic telemetry viewer.
- **Audit Logs (/admin/audit)**: Tamper-proof, searchable audit log of all clinical actions and data access events.

3.2 Hardware Interfaces

The firmware running on the ESP32 interacts with hardware components using standard bus protocols:

1. I2C Bus Protocol (Shared, 400 kHz Fast Mode):

- SDA on GPIO 21, SCL on GPIO 22.
- Master: ESP32. Slaves: SSD1306 Display (0x3c) and DS3231 RTC (0x68).

2. Synchronous Serial Interface (HX711 24-bit ADC):

- Clock line SCK on GPIO 18 driven by ESP32; Data line DOUT on GPIO 35 sampled by ESP32.
- Timing: Minimum 0.1 μ s high/low clock pulses; 25 pulses per conversion to select Channel A with 128 gain.

3. UART Serial Protocol (DFPlayer Mini):

- UART2 Hardware Serial: TX on GPIO 17, RX on GPIO 16. Baud rate: 9600 bps, 8 data bits, no parity, 1 stop bit.

4. PWM Pulse Generator (SG90 Servo):

- Frequency: $50 \times 10^3 \text{ Hz}$ ($20 \times 10^{-3} \text{ ms}$ period). Duty cycle: $1.0 \times 10^{-3} \text{ ms}$ (0° , gate closed) to $2.0 \times 10^{-3} \text{ ms}$ (90° , gate open).

5. Digital Half-Step Sequence (28BYJ-48 Stepper):

- 8-phase excitation cycle driving GPIOs 13, 12, 14, 27 sequentially:
 $\{1, 0, 0, 0\} \rightarrow \{1, 1, 0, 0\} \rightarrow \{0, 1, 0, 0\} \rightarrow \{0, 1, 1, 0\} \rightarrow \{0, 0, 1, 0\} \rightarrow \{0, 0, 1, 1\} \rightarrow \{0, 0, 0, 1\} \rightarrow \{1, 0, 0, 1\}$.

3.3 Software Interfaces & Third-Party Services

- 1. PostgreSQL Database:** Primary relational datastore (PostgreSQL 15+) using `psycopg2-binary` and connection pooling via PgBouncer.
- 2. Redis 7 In-Memory Store:** Dual-purpose deployment for Celery broker queues (Database 0) and Django ASGI channel layers/cache (Database 1).
- 3. Groq Cloud LLM Inference:** High-speed inference endpoint supporting `llama-3.3-70b-versatile` and `mixtral-8x7b-32768` with structured OpenAI-compatible tool calling.
- 4. Firebase Cloud Messaging (FCM HTTP v1):** Push notification delivery to Android, iOS, and Web PWAs with custom payload handling.
- 5. Telegram Bot API:** Full-duplex webhook interface receiving `/start`, `/verify`, and dose confirmation button callbacks.
- 6. Twilio REST API:** SMS dispatch and WhatsApp messaging fallback using E.164 phone formatting.
- 7. SendGrid Email API:** Transactional email delivery for onboarding, subscription invoices, and security alerts.
- 8. Razorpay & Stripe:** Dual payment gateways for automated recurring billing and hardware store purchases.
- 9. OpenFDA API:** National Library of Medicine drug interaction and pharmacovigilance database query interface.
- 10. ABDM Sandbox Gateway:** National Health Authority (NHA) API for ABHA address creation, demographic authentication, and health record linking.

3.4 Communication Protocols

- 1. RESTful HTTP/HTTPS API:** Standard JSON API compliant with OpenAPI 3.0 specification.
- 2. WebSocket Secure (WSS):** Asynchronous full-duplex communication over Daphne ASGI for live notifications, telemetry, and consultation signaling.
- 3. IoT Polling Protocol:** Low-overhead HTTP/1.1 REST polling mechanism:
 - Schedule Poll: `GET /api/v1/iot/devices/{id}/schedule/` every 60 seconds.
 - Command Poll: `GET /api/v1/iot/devices/{id}/commands/` every 10 seconds.
 - Heartbeat: `POST /api/v1/iot/devices/{id}/heartbeat/` every 5 minutes.
 - Dose Event: `POST /api/v1/iot/events/` immediately upon sensor trigger.
- 4. Device Authentication Header:** Every IoT request must carry X-Device-Key: `<secret_device_token>`. Unauthenticated requests are rejected with HTTP 401 Unauthorized.

4. System Features & Detailed Functional Requirements

4.1 Feature 1: Identity, Authentication & Role-Based Access Control (RBAC)

4.1.1 Description

Provides multi-tenant identity verification, multi-factor authentication (MFA), role-based permission scoping, and session security.

4.1.2 Functional Requirements

- **[FR-1.1] User Registration & Social Auth:** The system shall support user registration via email/password and Google OAuth2. Roles include: PATIENT, CAREGIVER, DOCTOR, PHARMACIST, TENANT_ADMIN, and SUPER_ADMIN.
- **[FR-1.2] JWT Token Lifecycle:** The system shall issue an HMAC-SHA256 signed JWT Access Token (lifetime: 15 minutes) and a Refresh Token (lifetime: 7 days) upon successful authentication.
- **[FR-1.3] Time-Based One-Time Password (TOTP) MFA:** The system shall provide RFC 6238 TOTP MFA for doctor and administrative accounts, generating standard QR code secrets.
- **[FR-1.4] Role-Based Route Guards:** The system shall enforce endpoint authorization via permission classes (IsPatient, IsCaregiver, IsDoctor, IsTenantAdmin, IsSuperAdmin).
- **[FR-1.5] Multi-Tenant Context Isolation:** The system shall resolve tenant context using subdomains (<tenant>.aarogyam.app) or the X-Tenant-ID header, filtering all database queries to the active tenant.

4.2 Feature 2: Smart IoT Pill Dispenser & Firmware Lifecycle

4.2.1 Description

Manages physical device registration, FreeRTOS state transitions, gravimetric intake validation, and remote hardware commands.

4.2.2 Functional Requirements

- **[FR-2.1] Device Pairing & Token Generation:** The system shall generate a cryptographically random 32-character pairing token for each dispenser. Scanning the device QR code via the patient portal links the hardware to the patient record.
- **[FR-2.2] FreeRTOS Dose Sequence Execution:**
 - *Step 1 (Schedule Trigger):* When local RTC matches schedule time, taskSensor enters STATE_DISPENSING, updates OLED with medicine name, beeps buzzer, and plays DFPlayer Track 1 ("Dawai lene ka waqt ho gaya").
 - *Step 2 (Compartment Rotation):* taskMotor acquires xMotorMutex and rotates stepper motor 1024 half-steps (90°) to align the scheduled compartment with the dispensing chute.
 - *Step 3 (Contactless Hand Detection):* HC-SR04 ultrasonic sensor continuously samples distance. When distance ≤ 15 cm is detected for $>500\text{ ms}$, servo actuates to 90° (open gate) and plays Track 2 ("Apni dawai nikaalo").
 - *Step 4 (Hand Withdrawal & Gate Closure):* When hand distance > 15 cm for $>1.0\text{ s}$, servo returns to 0° (closed gate).
 - *Step 5 (Gravimetric Weight Verification):* taskSensor enters STATE_WEIGHT_CHECK, settles for 3.0 s, and averages 10 readings from HX711. Weight delta is calculated:

$$\Delta W = W_{\text{before}} - W_{\text{after}}$$
 - If $\Delta W \geq \text{expected_weight} \times 0.85$: Status set to TAKEN. Audio Track 3 plays ("Shukriya, dawai le li gayi").
 - If $0 < \Delta W < \text{expected_weight} \times 0.85$: Status set to PARTIAL_DOSE.
 - If $\Delta W \leq 0$: Status set to DOSE_NOT_TAKEN.
 - *Step 6 (Event Transmission):* Device posts event payload to /api/v1/iot/events/:

```
json { "device_id": "ESP32-AAROGYAM-0042", "event_type": "DOSE_TAKEN", "compartment": 1, "weight_delta_grams": 0.45, "battery_level": 94, "timestamp": "2026-09-16T08:00:15Z" }
```
- **[FR-2.3] Missed Dose Lockdown:** If no hand is detected within 60 minutes of schedule trigger, the device enters STATE_GATE_LOCKED, locks the servo gate, sounds periodic alert beeps, plays Track 4 ("Dawai nahi li, caregiver ko alert kiya"), and dispatches DOSE_TIMEOUT to the backend.
- **[FR-2.4] Remote Gate Unlock Override:** A caregiver can issue an unlock command from the web portal. The backend queues an UNLOCK_LID command. Upon receipt, the dispenser plays Track 5 ("Dispenser unlock ho gaya") and re-enters STATE_DISPENSING for 15 minutes.
- **[FR-2.5] Compartment Refill Wizard:** Activating FILL_MODE rotates each compartment sequentially to the top opening, allowing the patient or caregiver to load tablets while weighing the initial compartment baseline.

4.3 Feature 3: Prescription & Scheduling Engine

4.3.1 Description

Maintains patient medication schedules, calculates dynamic reminder times, and coordinates dosage events.

4.3.2 Functional Requirements

- **[FR-3.1] Prescription Data Model:** Prescriptions must record medication name, dosage form (tablet, capsule, syrup), strength, route of administration, daily frequency, meal timing (BEFORE_MEAL, AFTER_MEAL, WITH_MEAL), start date, end date, total quantity, and remaining quantity.
- **[FR-3.2] Daily Reminder Generation:** A Celery Beat task running daily at 00:00 UTC evaluates all active prescriptions and instantiates daily Reminder records with specific target datetimes adjusted for the patient's local timezone.
- **[FR-3.3] Dose Escalation Flow:** When a scheduled reminder is unacknowledged:
 - $\$T + 0\text{min}$: Primary alert dispatched to IoT dispenser and In-App notification.
 - $\$T + 15\text{min}$: Mobile push notification (FCM) and Telegram bot reminder.
 - $\$T + 30\text{min}$: SMS notification to patient and primary caregiver.
 - $\$T + 60\text{min}$: Caregiver emergency escalation alert triggered, dispenser locked, and HIGH_RISK_DETECTED event fired.

4.4 Feature 4: AI Adherence Prediction & Explainability Engine

4.4.1 Description

Predicts patient non-adherence risk over a forward-looking 7-day window and explains contributing risk factors using machine learning.

4.4.2 Functional Requirements

- **[FR-4.1] Machine Learning Model Architecture:** The adherence risk model uses an XGBoost Classifier (adherence_risk_xgb_v1.9.0.pkl) trained on historical dose logs, feature-engineered variables, and clinical attributes.
- **[FR-4.2] Feature Vector Specification:**
 - adherence_rate_last_7d: Percentage of doses taken on time in past 7 days.
 - adherence_rate_last_30d: Percentage of doses taken on time in past 30 days.
 - current_streak_days: Consecutive days with 100% adherence.
 - missed_doses_last_14d: Total count of missed dose events in past 14 days.
 - timing_variance_minutes: Standard deviation of intake times relative to scheduled times.
 - regimen_complexity_score: Number of concurrent medications multiplied by daily dosing frequencies.
 - age: Patient chronological age.
 - is_living_alone: Boolean indicator of independent living.
- **[FR-4.3] Output Risk Classification:**
 - Risk Score: Normalized probability output (0 - 100).
 - Risk Level: LOW (0 - 35), MEDIUM (36 - 65), HIGH (66 - 100).
- **[FR-4.4] SHAP Explainability:** The inference engine applies shap.TreeExplainer to calculate local SHapley values for each inference, generating human-readable risk attribution strings (e.g., "Risk elevated due to high dosage timing variance (+18%) and living alone (+12%)").
- **[FR-4.5] Predictive Refill Consumption Forecasting:** Calculates the exact projected date of medication exhaustion based on moving-average daily consumption, triggering replenishment alerts 5 days prior to stock exhaustion.

4.5 Feature 5: Autonomous Agentic Runtime (Groq LLM Pipeline)

4.5.1 Description

An autonomous agentic reasoning pipeline built in `backend/apps/agent_runtime/` executing an Observe-Reason-Plan-Act-Evaluate loop for high-risk clinical interventions and auto-refills.

4.5.2 Functional Requirements

- **[FR-5.1] Event-Triggered Enqueue:** When `HIGH_RISK_DETECTED` or `REFILL_THRESHOLD_REACHED` events fire, the system spawns asynchronous Celery tasks (`evaluate_intervention_needed` or `evaluate_refill_needed`), decoupling LLM inference from synchronous API responses.
- **[FR-5.2] Observe Phase:** The agent compiles contextual memory from the database: patient demographics, recent dose logs, SHAP risk factors, active caregiver contact preferences, and prescription history.
- **[FR-5.3] Reason & Plan Phase:** The system constructs a structured prompt for the Groq LLM API (`llama-3.3-70b-versatile`) with strict JSON tool schemas. The LLM formulates a multi-step intervention plan:
 - Step 1: Query doctor availability or patient preference.
 - Step 2: Select high-priority communication channel.
 - Step 3: Compose personalized Hindi/English intervention message.
 - Step 4: Propose dispenser schedule adjustment or pharmacy refill order.
- **[FR-5.4] Policy Engine & Safety Guardrails:** All proposed actions pass through `PolicyEngine` before execution:
 - *Rule 1:* Autonomous medication dosage changes are strictly prohibited without verified doctor approval.
 - *Rule 2:* Maximum 1 automated phone call per patient per 24-hour cycle.
 - *Rule 3:* Pharmacy refill orders exceeding ₹5,000 require explicit caregiver/patient approval.
- **[FR-5.5] Tool Execution:** Approved actions are executed via `ToolRegistry`:
 - `send_multichannel_alert`: Dispatches alerts across selected channels.
 - `schedule_doctor_consultation`: Books an emergency tele-consultation slot.
 - `create_refill_order`: Places an automated order with the preferred pharmacy partner.
 - `lock_dispenser_gate`: Secures physical medication access.
- **[FR-5.6] Outcome Monitoring:** `OutcomeMonitor` tracks patient adherence for 72 hours post-intervention, recording intervention success metrics in `AgentAuditLog`.

4.6 Feature 6: Multi-Channel Real-Time Alert & Notification Dispatcher

4.6.1 Description

Delivers notifications across multiple communication channels based on recipient preferences and subscription plans.

4.6.2 Functional Requirements

- **[FR-6.1] In-App Real-Time WebSocket Delivery:** Pushes notification payloads over Django Channels to connected frontend clients within < 200 ms.
- **[FR-6.2] Firebase Cloud Messaging (FCM):** Sends mobile push notifications with custom priority tags, title, body, and deep-link routing URLs.
- **[FR-6.3] Telegram Interactive Bot (`apps.telegram_bot`):**
 - Direct account linking via phone number matching and 6-digit OTP verification.
 - Interactive reminder messages with inline buttons: [`Li (Taken)`], [`Nahi Li (Missed)`], [`15 min Baad (Snooze)`].
 - Inline button clicks update the `DoseLog` table and notify the IoT dispenser via `WebSocket/command` queue.
- **[FR-6.4] Twilio SMS & Voice Delivery:** Formats automated SMS reminders and triggers synthetic voice calls for critical high-risk alerts.
- **[FR-6.5] SendGrid Email Delivery:** Dispatches branded HTML email summaries, weekly adherence progress reports, and password reset tokens.

4.7 Feature 7: Caregiver Portal & Remote Tele-Intervention

4.7.1 Description

Empowers designated caregivers with monitoring, remote dispenser controls, and geofenced safety alerts.

4.7.2 Functional Requirements

- **[FR-7.1] Caregiver-Patient Link Management:** Supports 1-to-many and many-to-1 relationships with permission levels: `VIEW_ONLY`, `ALERTS_ONLY`, and `FULL_CONTROL`.
- **[FR-7.2] Remote Dispenser Locking & Unlocking:** Allows caregivers with `FULL_CONTROL` to remotely lock or unlock the physical dispenser gate.
- **[FR-7.3] Patient Geofence Configuration:** Caregivers can define a geographical coordinate and radius (e.g., 500 m around the patient's home). If the patient's mobile app leaves the boundary during a scheduled dose window, a `GEOFENCE_BREACH` alert is issued.

4.8 Feature 8: Doctor Clinical Portal & Adherence Auditing

4.8.1 Description

Provides clinical oversight, digital prescribing, and adherence monitoring for treating physicians.

4.8.2 Functional Requirements

- **[FR-8.1] Clinical Dashboard:** Displays an active patient census sorted by AI adherence risk score, highlighting non-compliant patients.
- **[FR-8.2] Digital Prescription Builder:** Enables physicians to generate structured prescriptions with dose timing, duration, and meal relations.
- **[FR-8.3] Real-Time Drug Interaction Checking:** Queries the OpenFDA database upon medication selection to alert the doctor of potential drug-drug or drug-disease contraindications.
- **[FR-8.4] Clinical PDF Report Generation:** Generates comprehensive adherence audit reports including graphs, dose timestamps, and vitals correlation using `xhtml2pdf`.

4.9 Feature 9: Autonomous Pharmacy Refill & Partner Integration

4.9.1 Description

Automates medication replenishment by bridging inventory tracking with external pharmacy fulfillment APIs.

4.9.2 Functional Requirements

- **[FR-9.1] Inventory Depletion Tracking:** Decrements `remaining_quantity` upon each verified `DOSE_TAKEN` event.
- **[FR-9.2] Refill Trigger:** When `remaining_quantity <= (dosage_per_day * refill_threshold_days)`, the system emits `REFILL_THRESHOLD_REACHED`.
- **[FR-9.3] Automated Partner Order Placement:** Generates a `RefillOrder`, calculates estimated cost, and invokes the partnered pharmacy API (`call_pharmacy_api`) with exponential backoff retries.
- **[FR-9.4] Delivery Replenishment Sync:** When the pharmacy marks the order as `DELIVERED`, the system updates `remaining_quantity` and schedules the dispenser refill wizard.

4.10 Feature 10: Gamification, Streaks & Behavioral Economics

4.10.1 Description

Incentivizes sustained adherence using habit-forming behavioral mechanisms.

4.10.2 Functional Requirements

- **[FR-10.1] Adherence Streaks:** Increments consecutive streak days when all daily scheduled doses are taken on time. Any missed dose resets the current streak to zero.
- **[FR-10.2] Milestone Badges:** Automatically awards digital achievement badges:
- `BRONZE_STREAK_7`: 7 consecutive days of 100% adherence.

- `SILVER_STREAK_30`: 30 consecutive days of 100% adherence.
- `GOLD_STREAK_90`: 90 consecutive days of 100% adherence.
- `CENTURION_100`: 100 total doses taken on time.
- `PERFECT_MONTH`: Zero missed or delayed doses in a calendar month.
- **[FR-10.3] Adherence Score Calculation**: Computes a dynamic composite score (0 - 1000) incorporating streak longevity, timing punctuality, and vitals recording frequency.

4.11 Feature 11: Vitals Monitoring & Telemetry Aggregation

4.11.1 Description

Correlates medication adherence with quantitative physiological biomarker improvements.

4.11.2 Functional Requirements

- **[FR-11.1] Supported Biomarkers**: Blood Pressure (Systolic/Diastolic in mmHg), Pulse (bpm), Blood Glucose (mg/dL with fasting/post-prandial tags), Weight (kg), SpO2 (%), and Body Temperature (°F/°C).
- **[FR-11.2] Clinical Thresholds & Anomaly Alerts**: Out-of-range vitals (e.g., Systolic > 160 mmHg or Glucose < 70 mg/dL) trigger instant alerts to caregivers and treating physicians.
- **[FR-11.3] Adherence-Vitals Correlation Visualizer**: Renders comparative trend graphs linking dose regularity with blood pressure/glucose stabilization over 30, 60, and 90-day intervals.

4.12 Feature 12: Healthcare Standards: ABHA (ABDM), FHIR R4 & Pharmacovigilance

4.12.1 Description

Aligns with national and international health data standards.

4.12.2 Functional Requirements

- **[FR-12.1] ABHA Account Integration**: Interfaces with India's ABDM Gateway to verify and link 14-digit ABHA numbers and `@abdm` health addresses.
- **[FR-12.2] HL7 FHIR R4 Serialization**: Exposes FHIR-compliant REST endpoints serializing patient adherence records as `FHIR MedicationStatement` and `MedicationAdministration` resources.
- **[FR-12.3] Pharmacovigilance & Adverse Event Logging**: Provides structured forms for patients and doctors to log adverse drug reactions (ADRs), automatically generating standard CIOMS / CDSCO-compatible pharmacovigilance reports.

4.13 Feature 13: Multi-Tenancy, Subscriptions, Hardware Store & Billing

4.13.1 Description

Manages commercial hospital tenant isolation, recurring SaaS subscriptions, and hardware purchases.

4.13.2 Functional Requirements

- **[FR-13.1] Tenant Data Isolation**: Enforces tenant scoping on all models via `TenantMiddleware`. Tenant data is inaccessible across organizational boundaries.
- **[FR-13.2] Subscription Tiers**:
 - `FREE`: 1 patient profile, basic push reminders, manual logging.
 - `CAREGIVER_PLUS`: IoT dispenser support, SMS/WhatsApp/Telegram alerts, 2 caregiver links.
 - `FAMILY_PRO`: Up to 5 patient profiles, vitals correlation, full AI risk scoring.
 - `HOSPITAL_ENTERPRISE`: Unlimited patients, multi-doctor dashboard, FHIR export, tenant branding.
- **[FR-13.3] Hardware Store & Inventory Fulfillment**: E-commerce catalog for purchasing smart dispensers and replacement carousels, with payment processing via Razorpay/Stripe.

5. Non-Functional Requirements (NFRs)

5.1 Performance Requirements

- **[NFR-1.1] API Response Latency:** 95% of standard HTTP REST API queries shall return responses in < 200 ms under a load of 1,000 concurrent requests.
- **[NFR-1.2] Real-Time Alert Delivery:** WebSocket and FCM push notifications shall be delivered to client devices in < 1.0 s from event trigger.
- **[NFR-1.3] IoT Edge Responsiveness:** The ESP32 FreeRTOS motor task shall initiate carousel rotation within < 100 ms of queue command dispatch.
- **[NFR-1.4] Weight Measurement Settling Time:** HX711 gravimetric verification shall stabilize and complete conversion within ≤ 3.0 s.
- **[NFR-1.5] LLM Inference Latency:** Autonomous agent Groq LLM tool planning shall complete within < 2.5 s.

5.2 Safety & Medical Failure-Mode Protection

- **[NFR-2.1] Physical Overdose Prevention:** The dispenser gate shall physically lock immediately after dose extraction, preventing double-dosing or premature compartment access.
- **[NFR-2.2] Jam & Misalignment Detection:** If the stepper motor fails to complete carousel rotation within 3.0 s, the firmware shall sound an alarm, lock the gate, and report a `MOTOR_STALL_ERROR` to the cloud.
- **[NFR-2.3] Fail-Safe Gate State:** In the event of total electrical power failure or system reset, the servo gate shall remain in the 0° mechanically closed position.
- **[NFR-2.4] Medical Advice Disclaimers:** All AI risk predictions and agentic nudges must display mandatory clinical disclaimers: *"For adherence assistance only; does not replace medical advice."*

5.3 Security Requirements

- **[NFR-3.1] Field-Level PHI Encryption:** Patient identifiers, medical conditions, and phone numbers shall be encrypted at rest in PostgreSQL using AES-128/256 via Fernet symmetric keys.
- **[NFR-3.2] Transport Layer Security:** All external communications (HTTPS, WSS) must enforce TLS 1.3 encryption.
- **[NFR-3.3] Hardware Mutual Authentication:** IoT dispensers shall authenticate each request using an immutable, cryptographically generated X-Device-Key.
- **[NFR-3.4] Immutable Audit Trail:** All reads, writes, and modifications to Protected Health Information (PHI) must append an immutable record to the `audit_log` table, recording user ID, IP address, timestamp, and action.

5.4 Reliability, Fault Tolerance & Offline Operation

- **[NFR-4.1] Offline Schedule Preservation:** The ESP32 shall cache up to 7 days of medication schedules in non-volatile flash memory (NVS/EEPROM). In the event of Wi-Fi loss, the device shall execute scheduled doses autonomously using the battery-backed DS3231 RTC.
- **[NFR-4.2] Offline Event Buffering:** Doses taken while offline shall be stored in local flash memory and synchronized in chronological order immediately upon Wi-Fi reconnection.
- **[NFR-4.3] Cloud High Availability:** The backend cloud infrastructure shall maintain $\geq 99.9\%$ operational uptime.

5.5 Regulatory & Compliance Requirements

- **[NFR-5.1] HIPAA & DISHA Alignment:** Meets Indian Digital Information Security in Healthcare Act (DISHA) standards and US HIPAA Security Rule standards for electronic protected health information (ePHI).
- **[NFR-5.2] Digital Personal Data Protection (DPDP) Act 2023:** Implements explicit user consent collection, data principal access controls, and "Right to be Forgotten" account erasure routines.

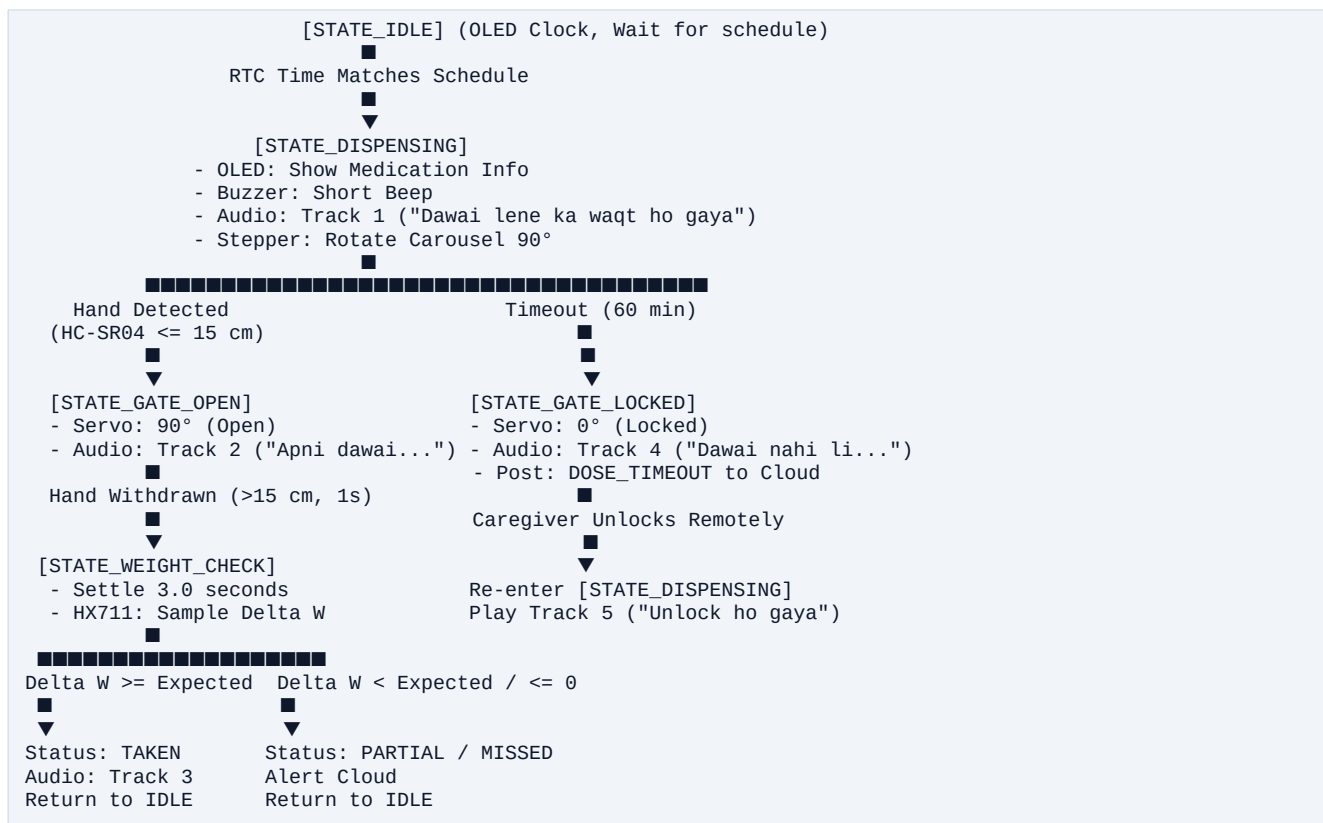
5.6 Usability, Accessibility & Localization

- **[NFR-6.1] Multilingual Audio Prompting:** The physical dispenser shall provide clear audio guidance in Hindi and English with adjustable volume (60 - 85 dB).
- **[NFR-6.2] Senior-Friendly UI Design:** The patient web interface shall adhere to WCAG 2.1 Level AA standards, featuring high-contrast color palettes (> 4.5:1), minimum font sizes of 16px, and intuitive touch targets (> 48x48 px).

6. System Architecture & Design Models

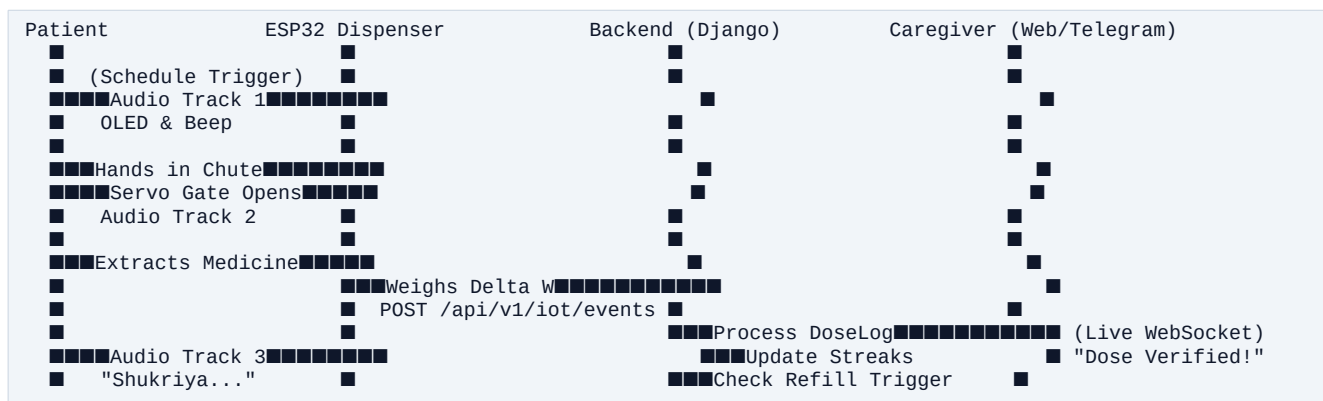
6.1 State Machine Diagrams

6.1.1 Physical Dispenser Dose Flow State Machine

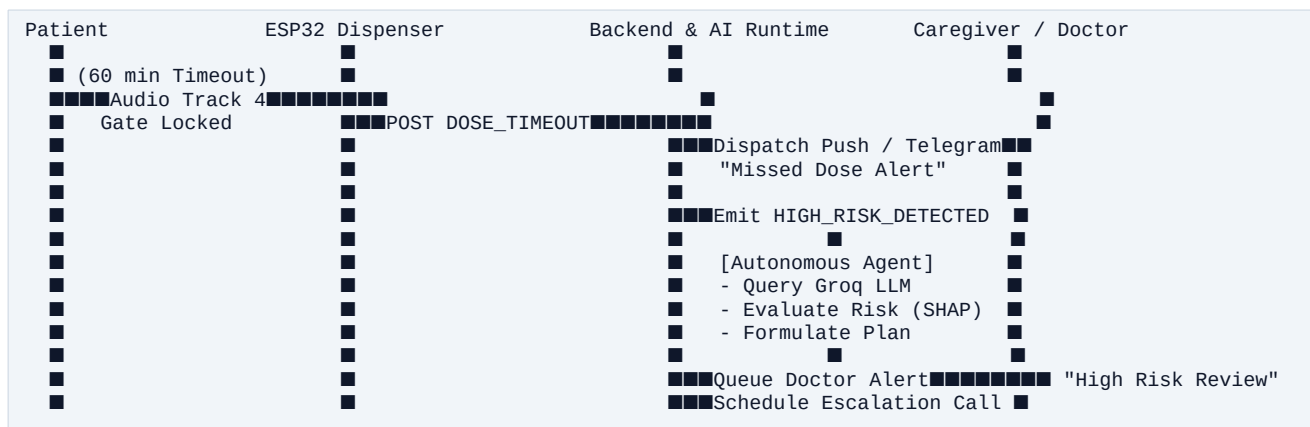


6.2 Sequence Diagrams

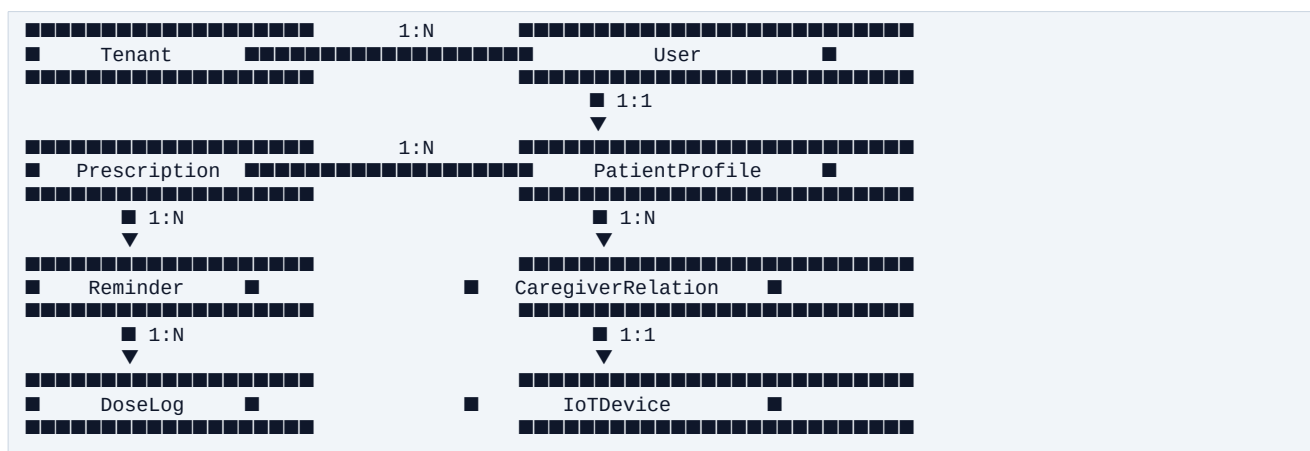
6.2.1 Dose Verification & Notification Sequence



6.2.2 Missed Dose & Autonomous Agent Escalation Sequence



6.3 Entity Relationship Model & Data Dictionaries



6.3.1 Core Database Tables

1. **users_user:**
2. **id:** UUID (Primary Key).
3. **email:** VARCHAR(255), Unique.
4. **phone_number:** VARCHAR(20), Unique.
5. **role:** VARCHAR(20) (PATIENT, CAREGIVER, DOCTOR, PHARMACIST, TENANT_ADMIN, SUPER_ADMIN).
6. **is_mfa_enabled:** BOOLEAN.
7. **mfa_secret:** VARCHAR(64) (Fernet Encrypted).
8. **iot_devices:**
9. **id:** UUID (Primary Key).
10. **device_serial:** VARCHAR(64), Unique.
11. **device_key_hash:** VARCHAR(128).
12. **patient_id:** UUID (Foreign Key $\rightarrow$ PatientProfile).
13. **status:** VARCHAR(20) (ONLINE, OFFLINE, LOCKED, ERROR).
14. **firmware_version:** VARCHAR(20).
15. **battery_level:** INTEGER.
16. **wifi_rssi:** INTEGER.
17. **last_heartbeat_at:** TIMESTAMP WITH TIME ZONE.
18. **scheduling_prescription:**
19. **id:** UUID (Primary Key).
20. **patient_id:** UUID (Foreign Key $\rightarrow$ PatientProfile).

21. `doctor_id`: UUID (Foreign Key $\rightarrow$ `DoctorProfile`, Nullable).
22. `medication_name`: VARCHAR(255).
23. `dosage_form`: VARCHAR(50).
24. `strength`: VARCHAR(50).
25. `dosage_per_day`: INTEGER.
26. `total_quantity`: INTEGER.
27. `remaining_quantity`: INTEGER.
28. `refill_threshold_days`: INTEGER (Default: 5).
29. `expected_dose_weight_grams`: DECIMAL(6, 3).
30. `start_date`: DATE.
31. `end_date`: DATE.
32. `status`: VARCHAR(20) (ACTIVE, COMPLETED, DISCONTINUED).
33. **scheduling_dose_log**:
34. `id`: UUID (Primary Key).
35. `prescription_id`: UUID (Foreign Key $\rightarrow$ `Prescription`).
36. `patient_id`: UUID (Foreign Key $\rightarrow$ `PatientProfile`).
37. `scheduled_time`: TIMESTAMP WITH TIME ZONE.
38. `actual_time`: TIMESTAMP WITH TIME ZONE.
39. `status`: VARCHAR(20) (TAKEN, MISSED, PARTIAL, SNOOZED).
40. `log_source`: VARCHAR(20) (IOT_DEVICE, MOBILE_APP, TELEGRAM_BOT, CAREGIVER_OVERRIDE).
41. `weight_delta_grams`: DECIMAL(6, 3).
42. **agent_runtime_agentgoal**:
43. `id`: UUID (Primary Key).
44. `patient_id`: UUID (Foreign Key $\rightarrow$ `PatientProfile`).
45. `goal_type`: VARCHAR(50) (ADHERENCE_INTERVENTION, AUTO_REFILL).
46. `trigger_event`: VARCHAR(50).
47. `status`: VARCHAR(20) (OBSERVE, REASON, PLAN, ACT, EVALUATE, COMPLETED, FAILED).
48. `reasoning_trace`: JSONB.
49. `created_at`: TIMESTAMP WITH TIME ZONE.

7. Verification, Testing & Acceptance Criteria

7.1 Test Strategy Matrix

Level	Scope	Tools / Frameworks	Target Metric
Unit Testing (Backend)	Models, Serializers, Handover logic, Policy rules	Pytest, Django Test Runner	>85% Line Coverage
Unit Testing (Frontend)	React components, Zustand stores, hooks	Vitest, React Testing Library	>80% Component Coverage
Integration Testing	REST APIs, Django Channels WebSocket consumers	Pytest-Django, Channels LiveServer	100% Core Endpoints Passing

Hardware-in-the-Loop	ESP32 Firmware FreeRTOS tasks, sensors, actuators	PlatformIO, Saleae Logic Analyzer	0 Stepper Skips, Accurate ΔW
End-to-End System	Complete dose flow: Schedule → Dispense → Verify	Playwright (Web), Mock ESP32 Client	End-to-end latency < 2 s

7.2 Hardware-in-the-Loop (HIL) Test Protocol

- Gravimetric Precision Test:**
- Apply standard calibration weights (0.25 g, 0.50 g, 1.00 g, 5.00 g) to the HX711 scale plate.
- Measure 50 consecutive weight samples.
- Acceptance Criterion:* Mean error must remain within $\leq \pm 0.05$ g.
- Carousel Rotation Endurance Test:**
- Execute 1,000 continuous 90° rotation cycles using the 28BYJ-48 stepper motor under nominal payload.
- Acceptance Criterion:* Zero mechanical stalls; compartment optical alignment variance $\leq \pm 2^\circ$.
- Power Loss & Recovery Protocol:**
- Interrupt electrical power during STATE_DISPENSING.
- Restore power after 60 seconds.
- Acceptance Criterion:* Device must read DS3231 RTC, query local NVS flash schedules, restore the servo gate to 0° (closed), and resume normal operational polling.

8. Appendices

8.1 Agent Handover Event Catalog

The system operates an asynchronous event bus connecting 28 specialized agents:

Event Identifier	Source Agent	Target Agent(s)	Payload Summary
DOSE_LOGGED	IoTAgent / TelegramBot	AdherenceAgent, AnalyticsAgent	patient_id, prescription_id, status, weight_delta
DOSE_MISSED	ReminderAgent	CaregiverAgent, NotificationAgent	patient_id, reminder_id, minutes_overdue
HIGH_RISK_DETECTED	AIAgent (XGBoost)	AgentRuntime, DoctorAgent, CaregiverAgent	patient_id, risk_score, shap_factors, trace_id
REFILL_THRESHOLD_REACHED	MedicationAgent	AgentRuntime, PharmacyAgent	prescription_id, remaining_quantity, days_left
REFILL_ORDER_PLACED	PharmacyAgent	NotificationAgent, StoreAgent	order_id, partner_id, medication_name, total_amount
DOCTOR_ALERT_TRIGGERED	DoctorAgent	NotificationAgent	doctor_id, patient_id, severity, clinical_reasons

GEOFENCE_BREACH	GeofenceAgent	CaregiverAgent, NotificationAgent	patient_id, current_lat, current_lon, distance_breached
DEVICE_TAMPER_DETECTED	IoTAgent	CaregiverAgent, AuditAgent	device_id, accelerometer_delta, timestamp

8.2 Audio Guidance Prompts Directory

Track #	Audio File	Hindi Voice Prompt	English Equivalent	Functional Trigger
01	0001.mp3	"Dawai lene ka waqt ho gaya hai"	"It is time to take your medicine"	Scheduled reminder time reached
02	0002.mp3	"Kripya apni dawai nikaalein"	"Please take out your medicine"	Dispenser gate opened (90°)
03	0003.mp3	"Shukriya, aapki dawai le li gayi hai"	"Thank you, your dose is confirmed"	Load cell verified dose extraction
04	0004.mp3	"Dawai nahi li gayi, caregiver ko soochit kiya gaya hai"	"Dose missed, caregiver has been alerted"	60-minute timeout expired
05	0005.mp3	"Dispenser unlock ho gaya hai"	"Dispenser has been unlocked"	Remote caregiver unlock command

End of Software Requirements Specification (SRS)